

	University of Central Punjab Faculty of Information Technology & Computer Sciences Data Structures, Spring 2026 Quiz 4 V2	
	Total Marks: 20	Time Allowed: 30 min
Section	Student Name	Student Registration
D5		

CLO #	CLO Statement	Taxonomy Level	PLO
3	Analyze algorithms to understand their structure and functionality.	C5	3

Question # 1: Product IDs are stored in a BST.

Write a program to:

- Insert product IDs.
- Find the kth smallest product ID.
- Count products greater than a given ID.
- Print the path to a searched product.

Question # 2: Insert:

70, 55, 60, 42, 83, 90, 93, 101, 65, 56

Tasks:

1. Draw the self-balancing tree after each insertion.
2. Identify the node(s) that violate the balancing properties of the tree.
3. Explain how the tree restores balance after each violation.
4. Specify the balancing operation performed (rotation and/or recolouring, where applicable).
5. Track all balancing operations throughout the insertion process.
6. Draw the final balanced tree.